Code:

```
void setup()(
pIr#mode(5, OUTPUT):
pirfiode (6, OUTPUT):
Serial. begin(9600);
)
void 100p () (
Serial, printin(ON): digitallrite(5,HIGH ):
digitallrite(6, LOw)
delay (1000);
SenJa1.prIntZn(*OFF"):
digitallrite (5, 0);
digitalkrite(6, 1);
de lay (1000);
)
```

Description:

The ESP32 microcontroller is used in this project to interface with sensors and show real-time pressure, temperature, and humidity values. The system's purpose is to show how embedded systems integrate hardware and software. Sensor data is processed by the ESP32 before being shown on the Serial Monitor.

Environmental data is gathered by the sensor (a BME280 in this instance), which sends it to the microcontroller for processing. This is perfect for uses like smart home systems and environmental monitoring. Continuous monitoring is provided by updating the sensor readings every two seconds.

By adding additional sensors or components, the modular architecture makes expansion simple. Because of its affordability, dependability, and energy efficiency, the system can be used in a variety of embedded applications. The Arduino IDE was used to write the code, guaranteeing interoperability with a variety of systems and devices. It is also easily adaptable to new features or different kinds of sensors.